

Enhancement Two Narrative

1. Briefly describe the artifact. What is it? When was it created?

This second enhancement for Algorithms and Data Structures is probably the smallest enhancement I am making to the project. This enhancement is where I am adding the ability for users to search by location and sort based on location, pricing, etc. For this enhancement I did not have to create new files. I only modified several existing files. The main work for this enhancement was the algorithms needed for handling the location. There are still areas for improvement and enhancements related to the location sorting, especially from a UX perspective.

Here are the new files created:

Backend:

- N/A

Frontend:

- N/A

Here are the modified files:

Backend:

- communityController.js – added haversine function that returns distances from the search location to facilitate location sorting

Frontend:

- App.jsx – this is where much of the enhancement work took place. Added functions haversine, minPrice, maxPrice, applyLocation, and clearLocation, as well as constants for location and sorting, namely the constant sortedCommunities which is a list of the communities sorted by distance from the search location. I also had to enhance the html creation code to add these features to the UI.
- App.css – I have mostly not touched the css code from the AI generated starter code. I think UX enhancements should come towards the end of the project, but there were quite a few enhancements to the css for the location and sorting features.
- FilterPanel.css – minor update to add the location feature to the filter panel

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

For the Algorithm and Data Structures enhancement I chose to add searching and sorting by location to my web application. The original code created by the generative AI just showed all the communities on the map. At first, it was not noticeable because the communities that have been seeded into the database (by the AI) for development were all in one general location. I chose this improvement for the second enhancement because I had to add new functions for the location, particularly the algorithm for sorting locations based off their distance from a searchable location (i.e. city, state, or zip code).

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

The focus of my second enhancement is the algorithms used for the location search functions. The course outcome most related would be “Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution, while managing the trade-offs involved in design choices.” I was unsure of how to implement the algorithm so, following my theme for this project, I used AI tools to generate a function that returns the distance between two points, one being the original “search” location, and the second point being other locations. I tested this function in a sandbox environment to make sure it was calculating the distances correctly and returning values I could work with. From there I just called this function as I iterated over the communities. Overall, the feature is working well currently. However, I have concerns about the algorithms performance as the database of communities increases. I do not want it to iterate over the entire database. I think I may take a second pass at this location search algorithm after I seed the database with more communities.

Insights after finishing the project:

Collaborating in a team environment – added comments throughout to make the code easier to understand and maintain.

Communicating with stakeholders – searching and sorting by location is a critical feature for customers.

Data Structures and algorithms – the core of this enhancement is the new algorithm to calculate the distances and sort by distance from the desired location.

Software engineering and database – required updating the filtering and sorting logic.

Security – validates the filtering inputs including the location.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

On many levels, this second enhancement was far easier than my first one. The first enhancement involved adding multiple new classes to both the frontend and backend, as well as adding a new model to the database. For the second enhancement I only modified a few files. That being said, the algorithm logic was challenging to get working right. The first enhancement, even though it seemed like a lot more work, was all things I have done before in classes and in personal projects. There was less learning curve for the first enhancement. The second enhancement really took some lower-level testing and debugging, trying to get the specific functions to work and work well together. There was a lot of focus on the distance function and the sorting by distance algorithm to get the process to behave correctly. This development process reminded me of problems I face at work where I am trying to test a very specific but complex requirement. On the other hand, the first enhancement was more modular development process, where I implement one little piece of functionality and test it, then move on to the next step.